

LAB 01 (Creating Github Account)

AYESHA MAQSOOD AWAN

FA24B1-SE-070

CLOUD COMPUTING

Section B

ayeshamaqsoodawan

Task 1: Create and Configure your github account

Email*

ayesha.awan95406@gmail.com



Password*

.....



Password should be at least 15 characters OR at least 8 characters including a number and a lowercase letter.

Username*

ayeshamaqsoodawan



Username may only contain alphanumeric characters or single hyphens, and cannot begin or end with a hyphen.

Your Country/Region*

Pakistan



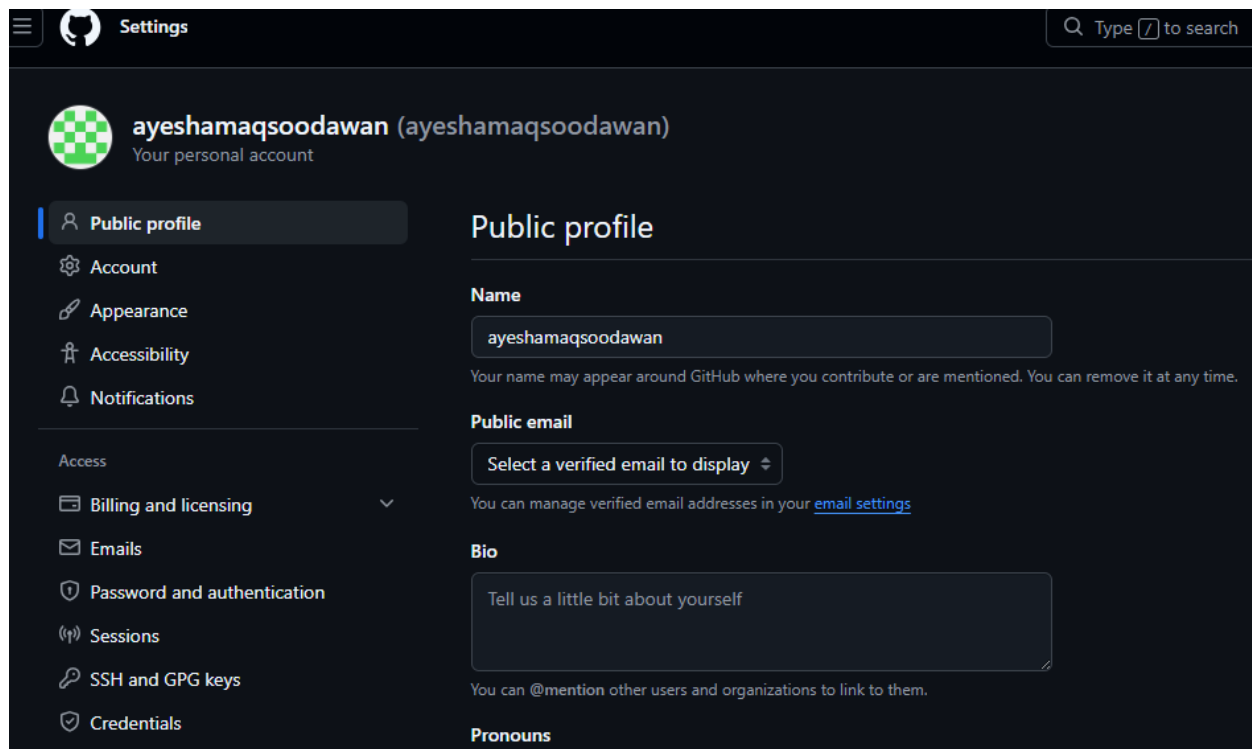
For compliance reasons, we're required to collect country information to send you occasional updates and announcements.

GitHub Copilot

- ☒ Sign up for Copilot Free for coding assistance on the web, in the IDE, and in the CLI. [See terms](#)

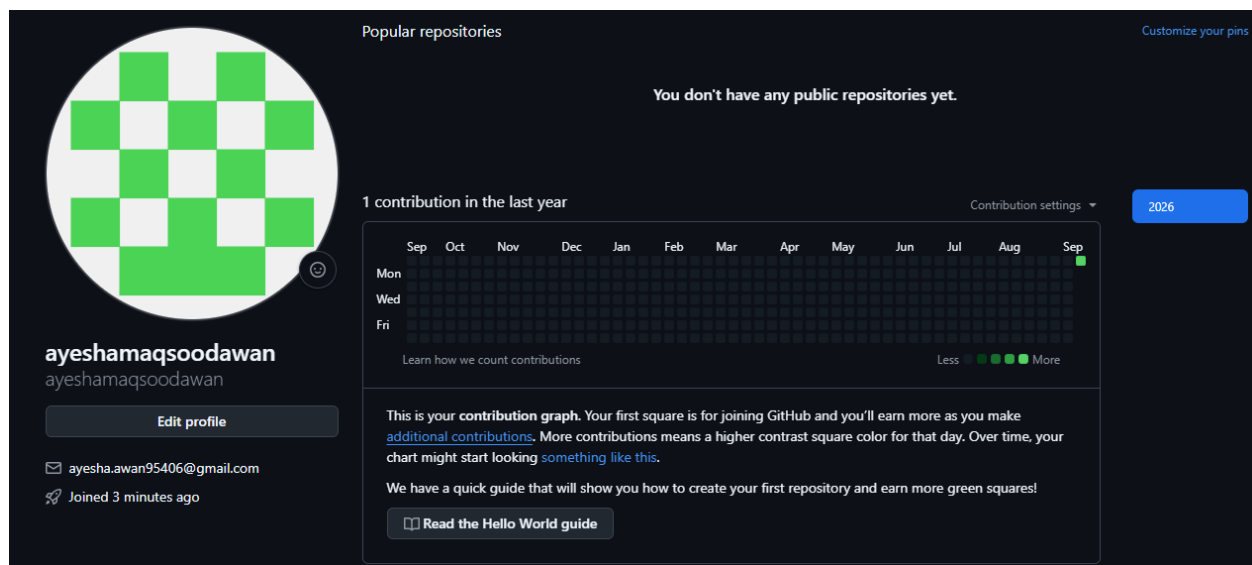
github_ayeshamaqsoodawan.png

Screenshot 1:



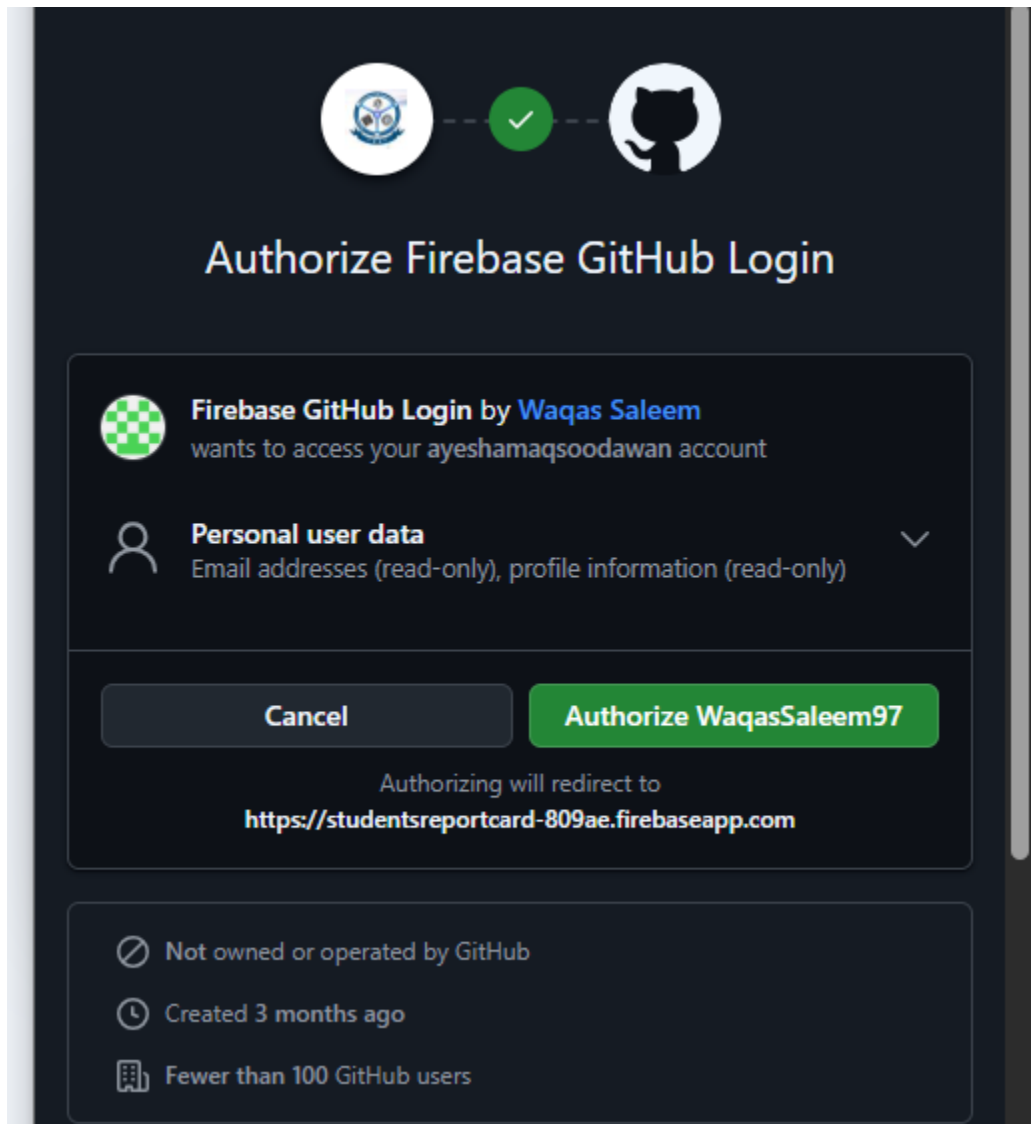
GitHub profile showing my profile name and username

Screenshot 2:



github_ayeshamaqsoodawan.png

Task 2: Register in student marks portal



Portal_login_page.png

Screenshot 1:



Complete registration

Enter your registration number, course and section.
The enrollment requires approval.

Registration number

Allowed: 2024-BSE-01 to 2024-BSE-99.

Course

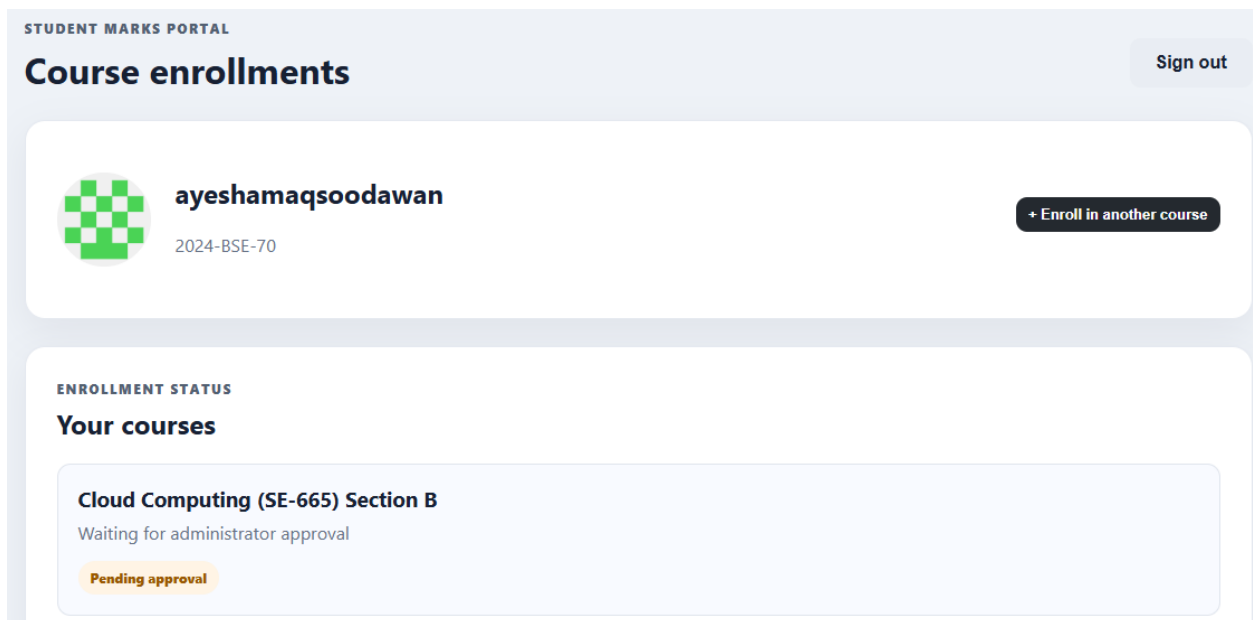
Section

Submit enrollment

Sign out

Portal_github_profile.png

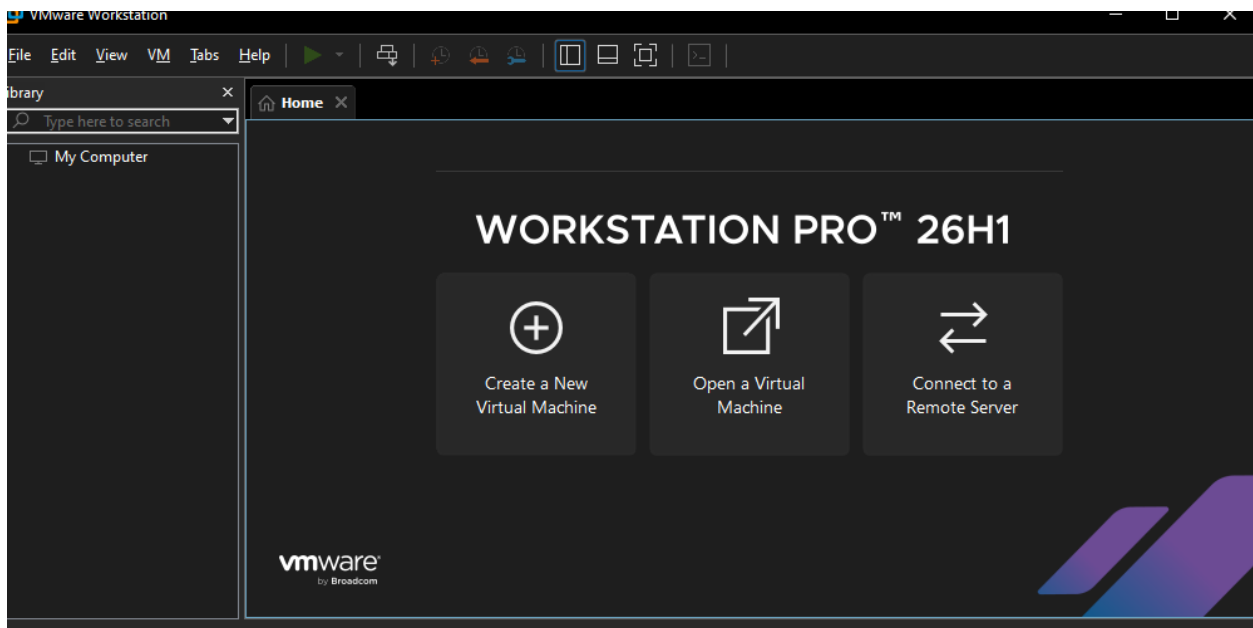
Screenshot 2/3:



Portal_enrollment_submitted.png

Task 3: Download VMware Workstation Pro and Ubuntu Server

Screenshot 1:



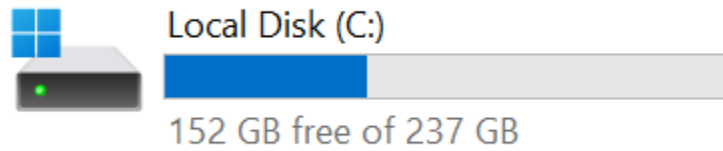
vmware_workstation.png

Screenshot 2:

 ubuntu-26.04.1-live-server-amd64	9/13/2026 8:32 AM	Disc Image File	2,859,240 KB
--	-------------------	-----------------	--------------

Ubuntu server ISO filename.

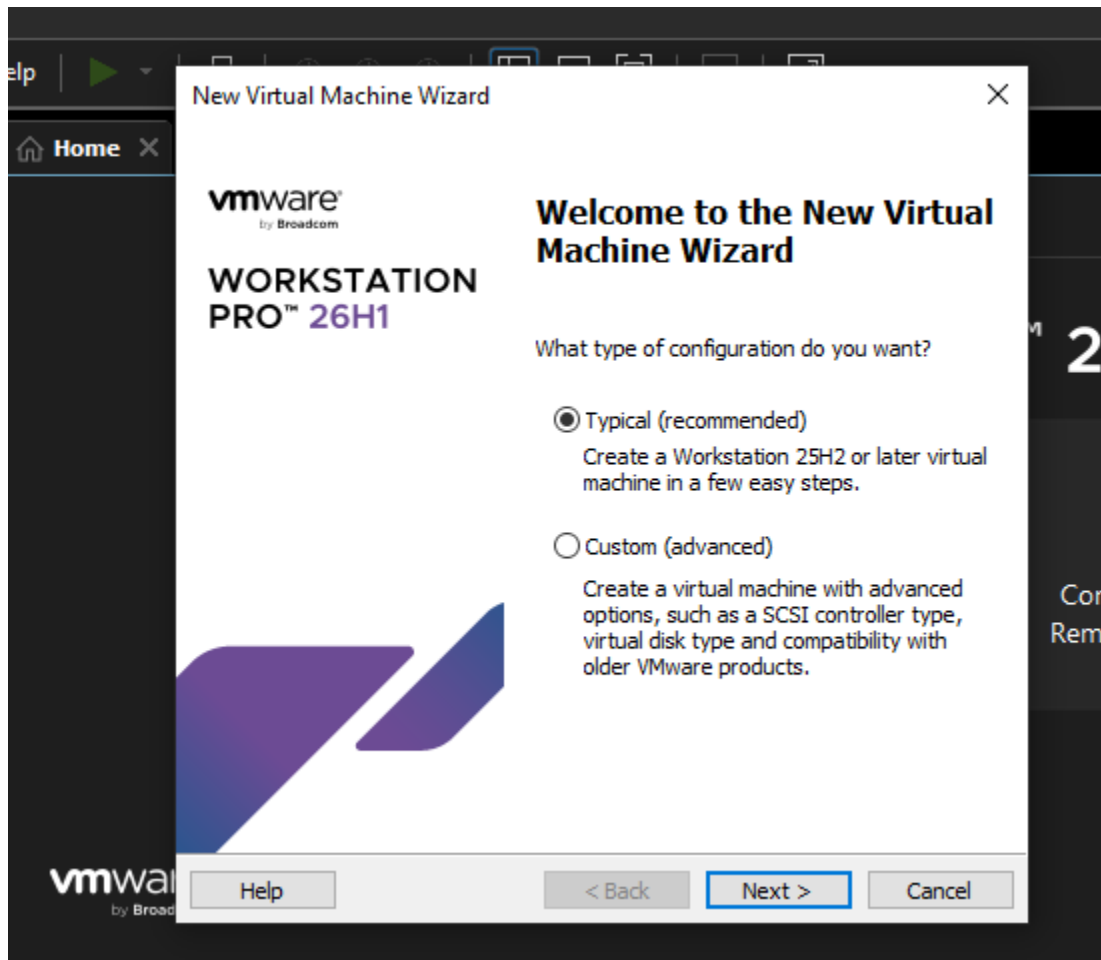
Screenshot 3:



Available storage showing atleast 20 GB free space.

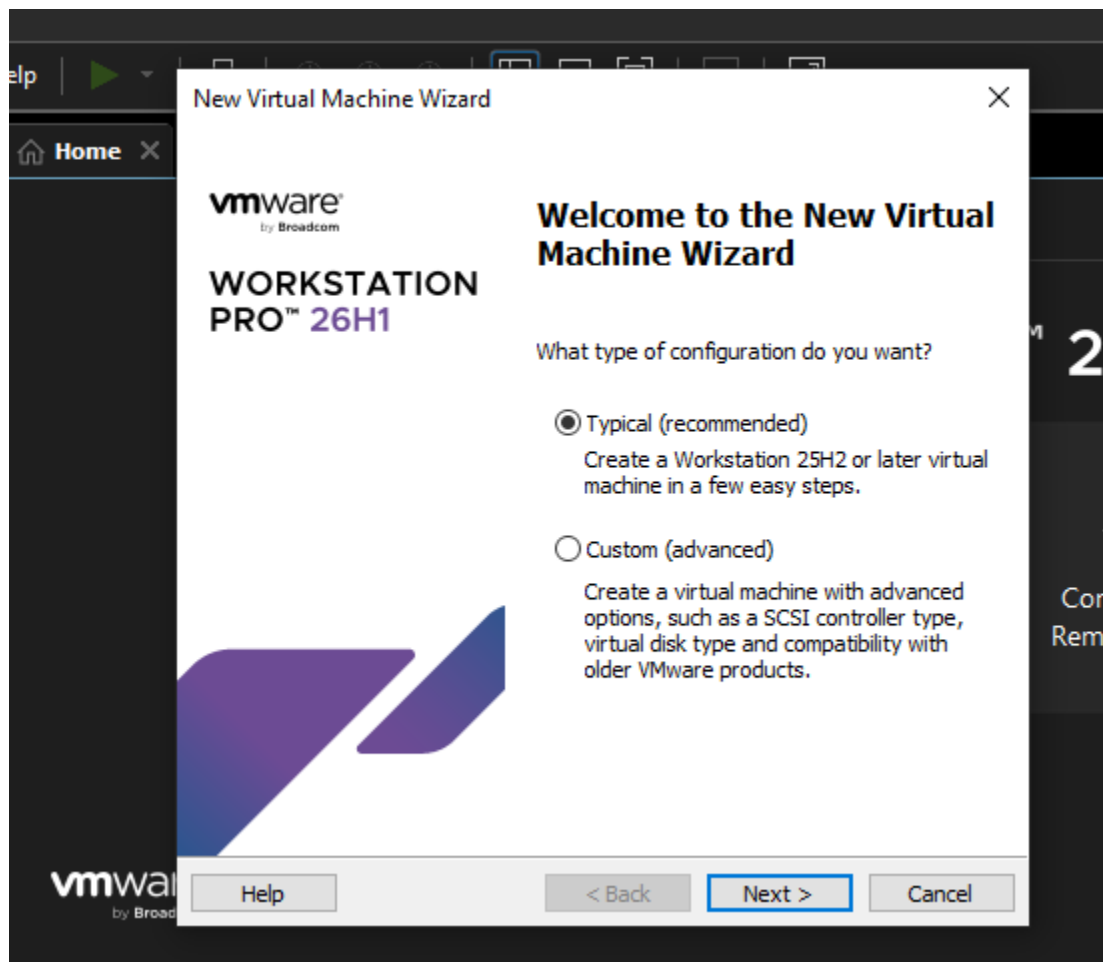
Task 4: Create the ubuntu server virtual machine

Screenshot 1: New VM Wizard



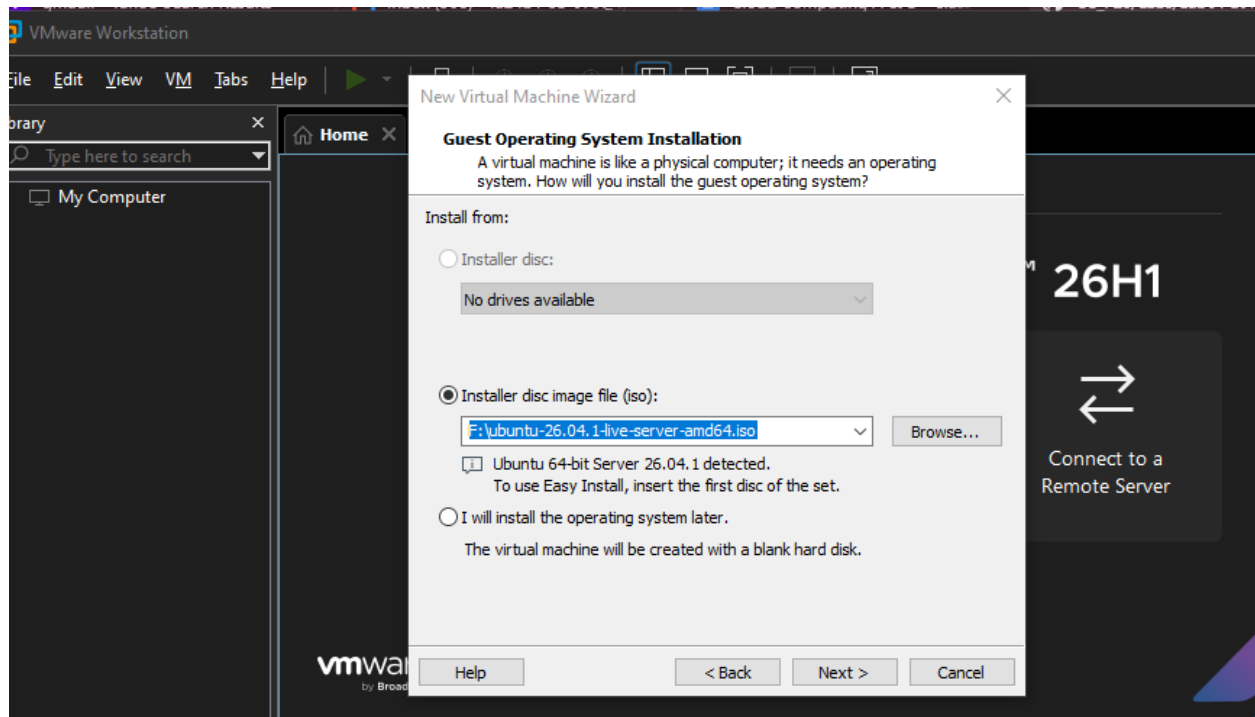
vm_creation_wizard.png

Screenshot 2: Typical configuration

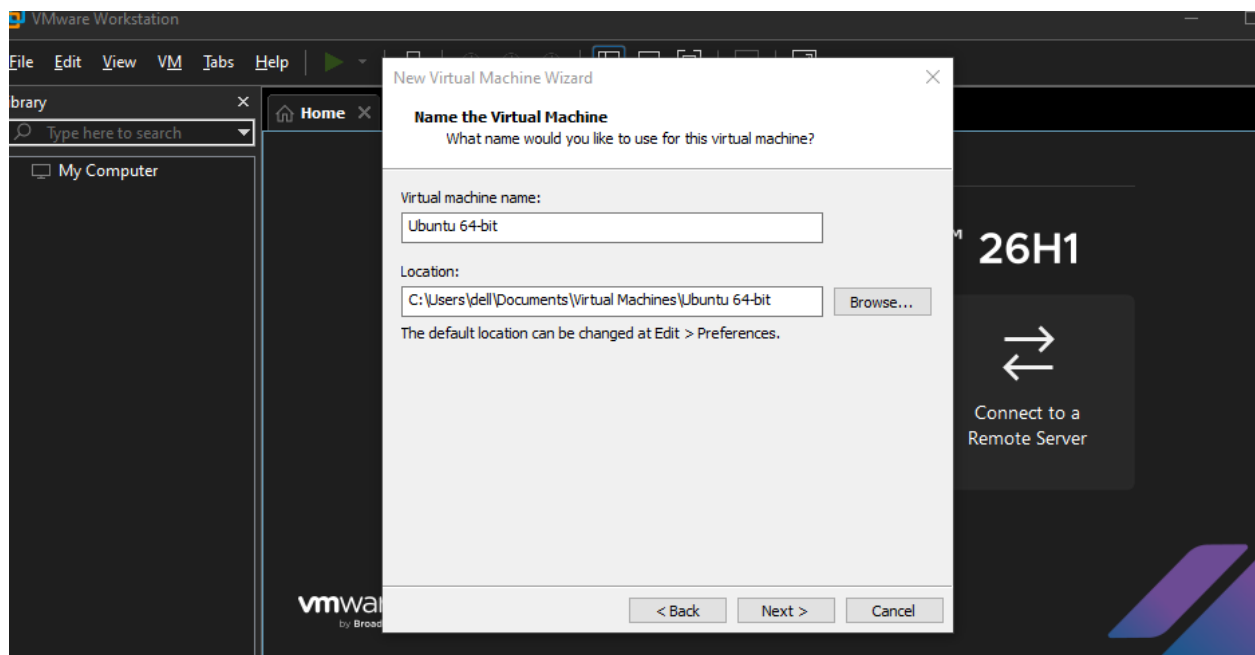


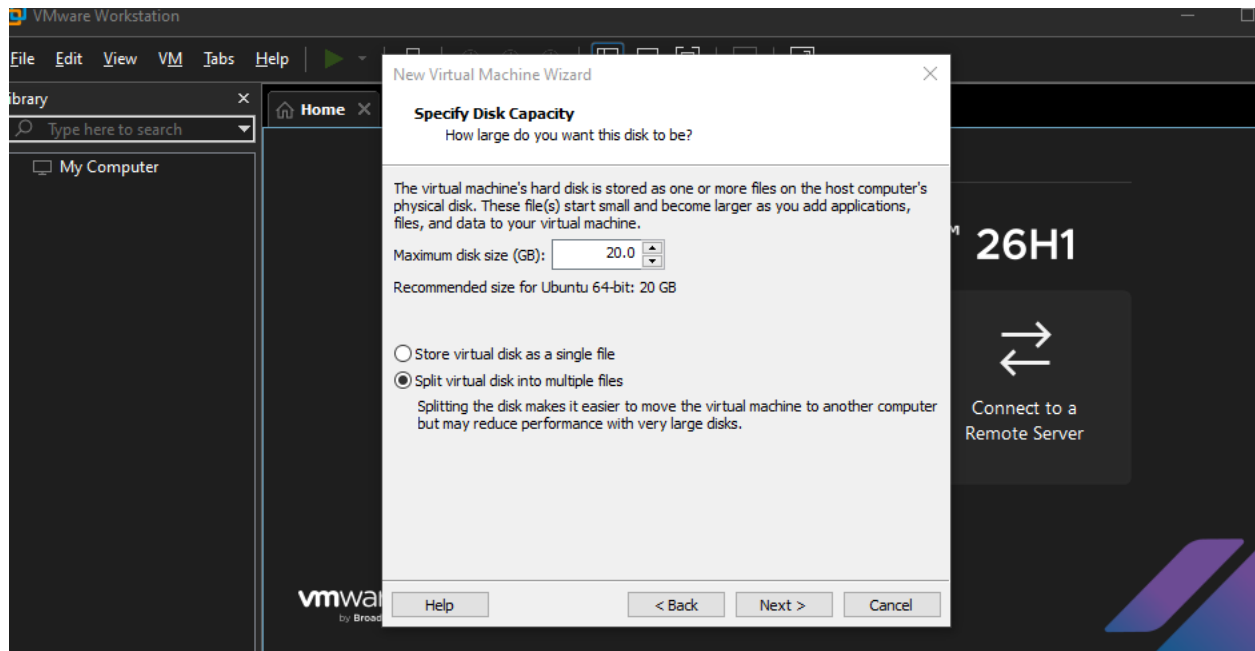
vm_typical_configuration.png

Screenshot 3:Ubuntu ISO selected

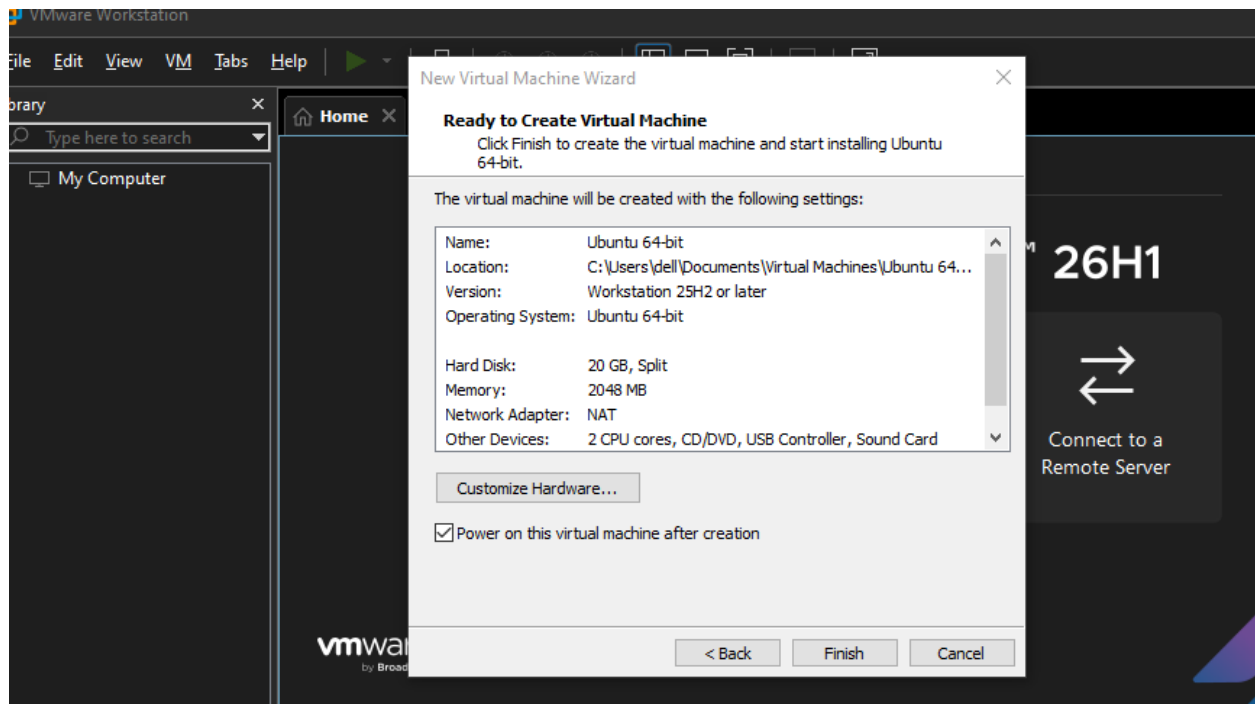


vm_ubuntu_iso_selected.png



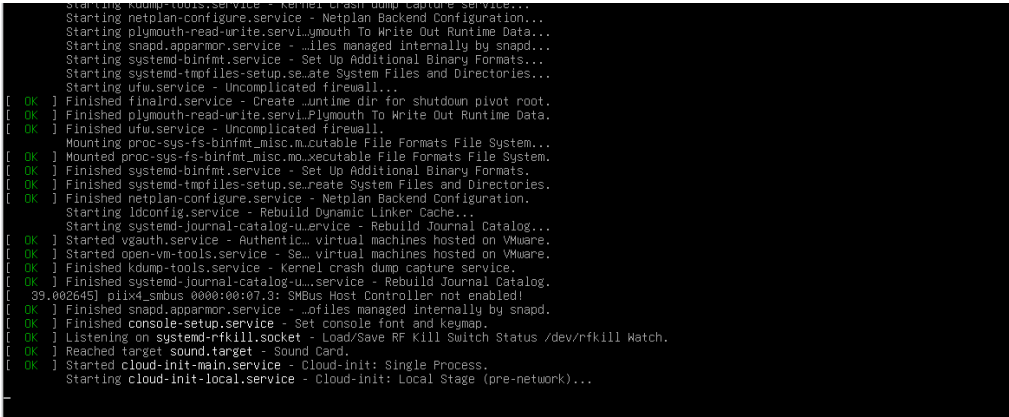


Screenshot 4: VM configuration summary



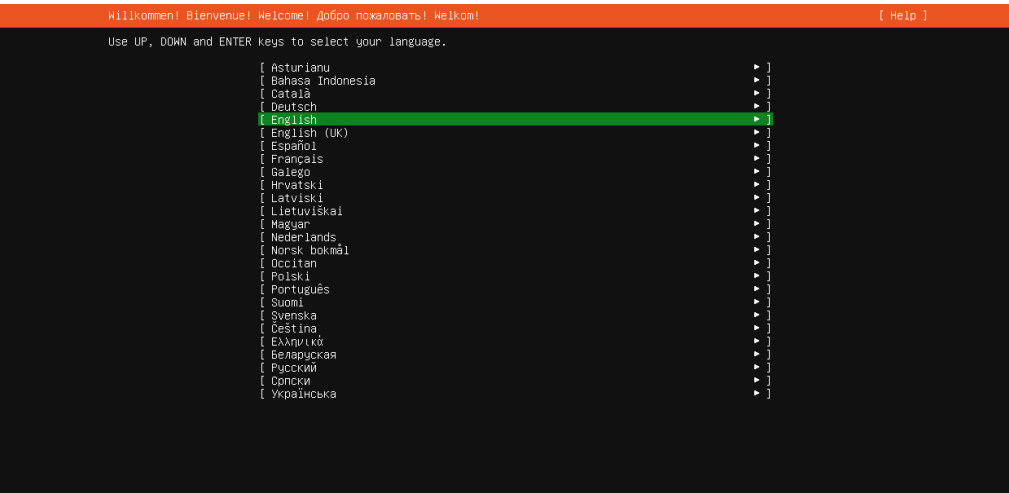
vm_configuration_summary.png

Screenshot 5:Ubuntu boot server



ubuntu_installer_boot.png

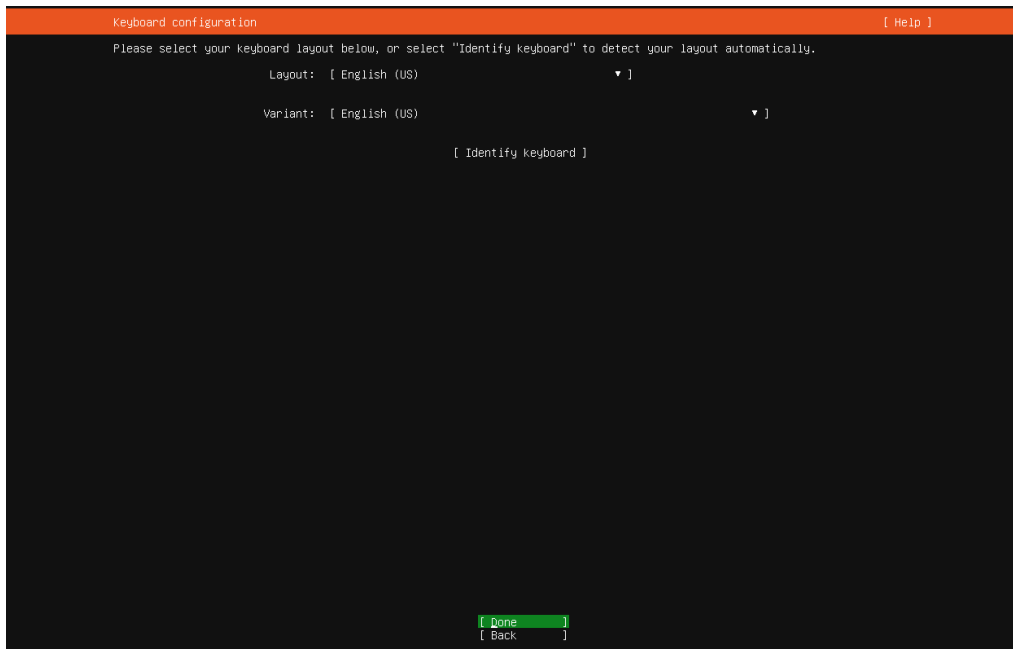
Task 5: Choose the Installation Language.



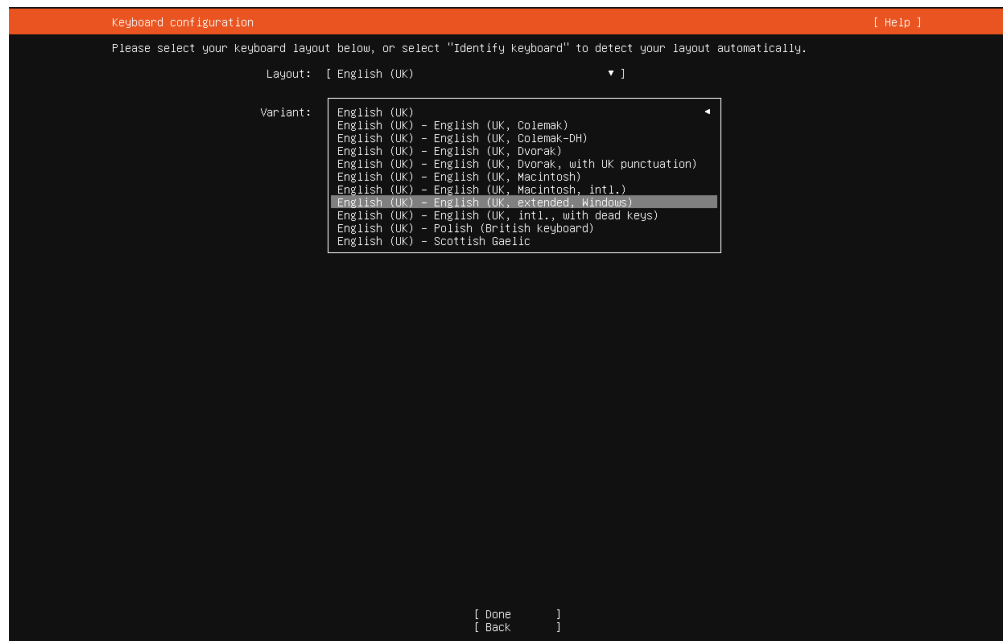
ubuntu_language.png

Task 6:Configure keyboard layout

Screenshot 1:



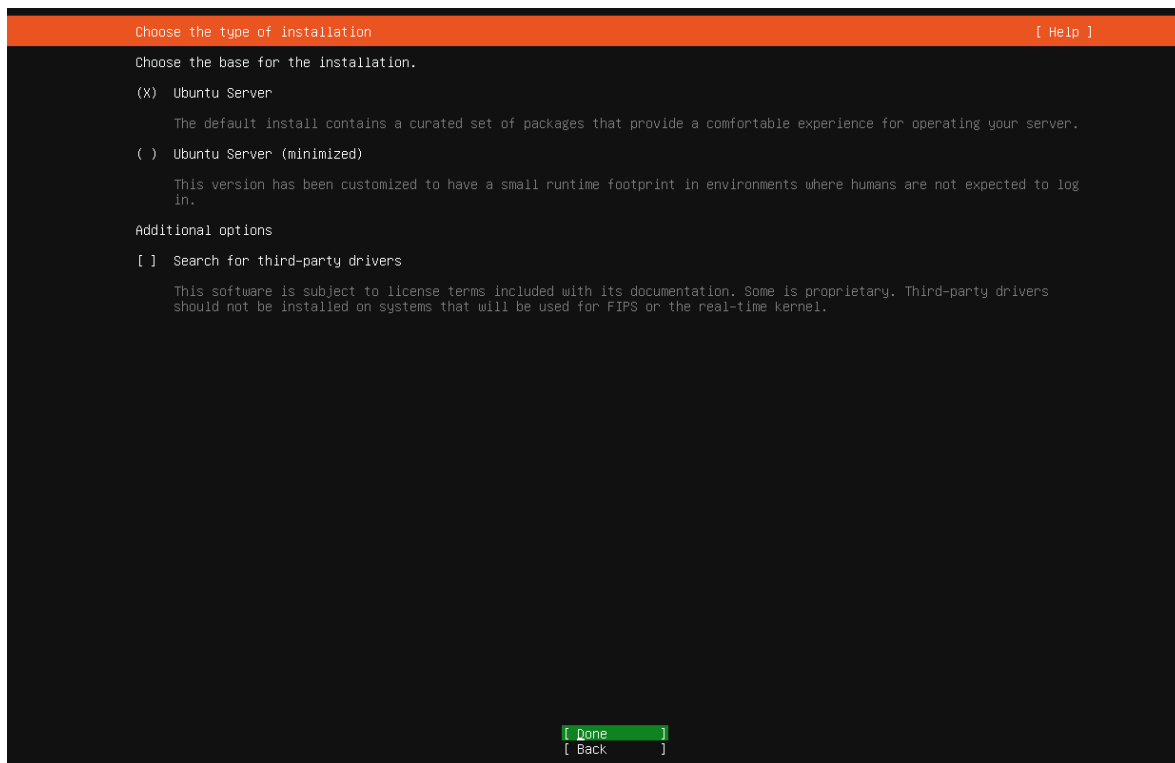
Screenshot 2:



ubuntu_keyboard_variant.png

Task 7: Choose the Ubuntu Server Installation Type

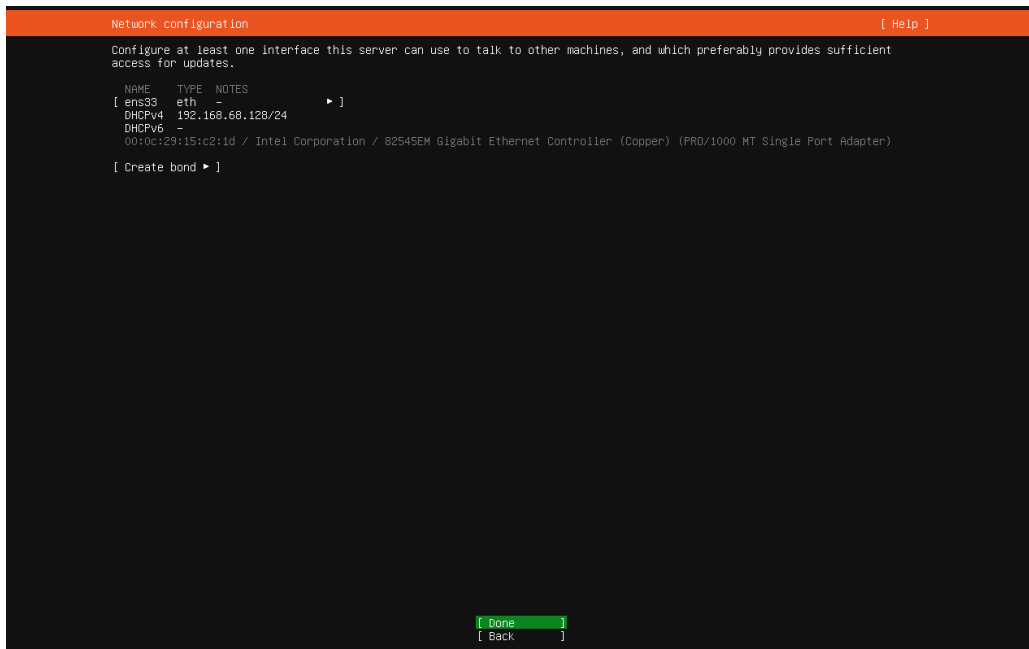
Screenshot 1:



Normal Ubuntu Server installation option selected.

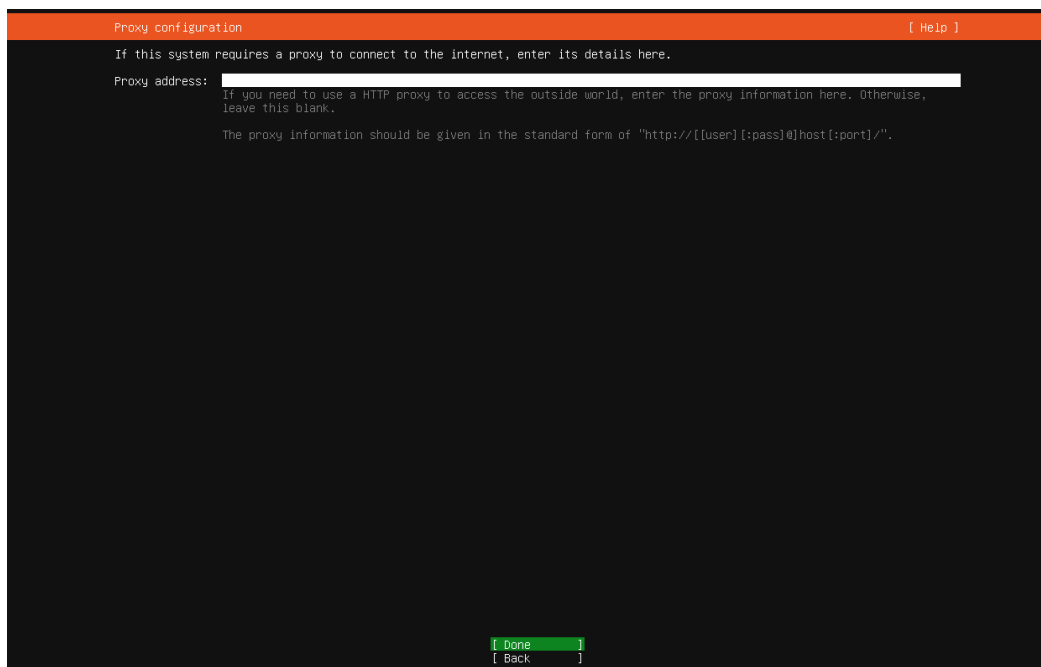
Task 8: Configure the network

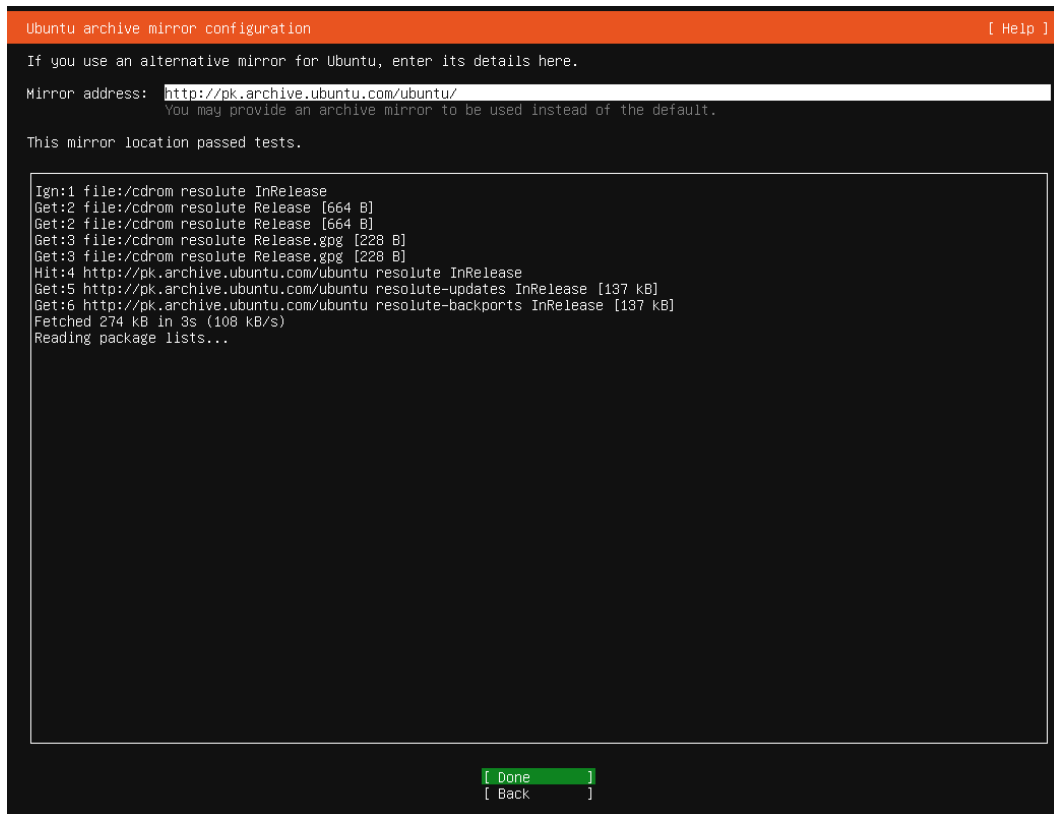
Screenshot 1:



Screenshot 2:

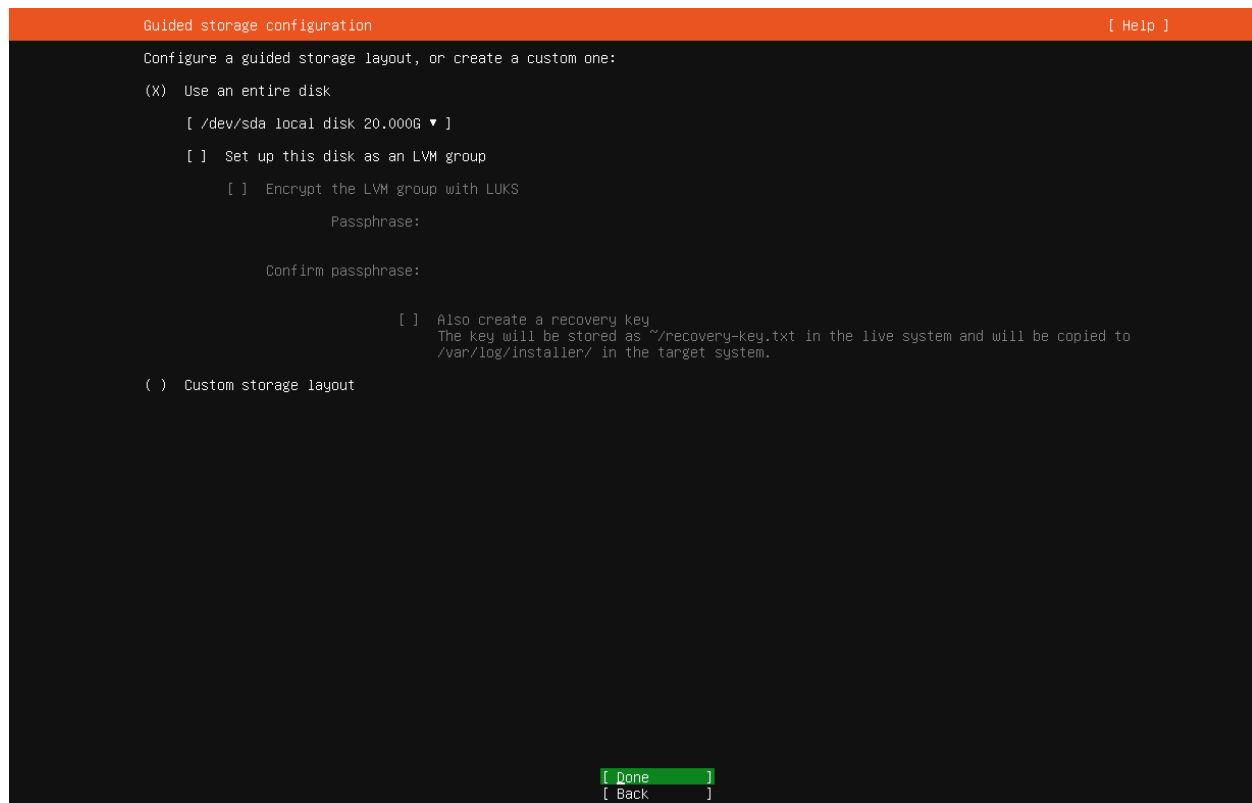
```
DHCPv4 192.168.68.128/24
DHCPv6 -
00:0c:29:15:c2:1d / Intel Corporation / 82545EM Gigabit Ethernet Controller (Copper) (PRO/1000 MT Single Port Adapter)
```



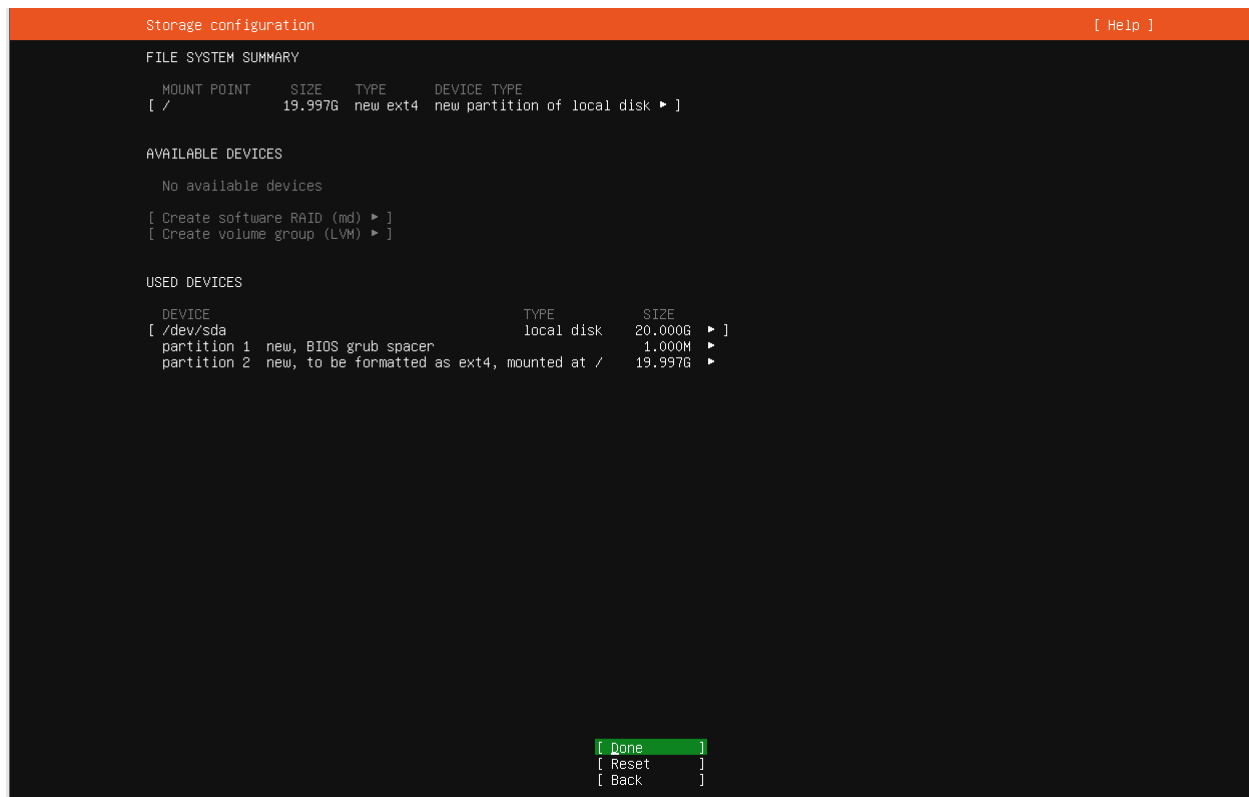


Task 9: Configure Storage

Screenshot 1:



Screenshot 2:



Task 10: Select the installation Disk

Screenshot 1:

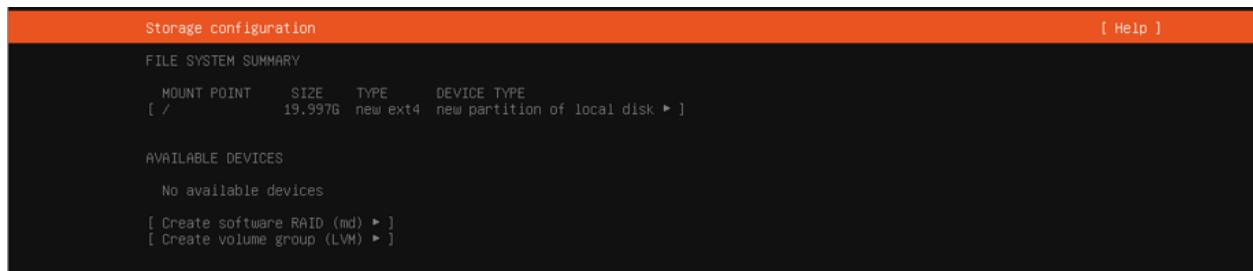
DEVICE	TYPE	SIZE
[/dev/sda	local disk	20.000G ▶]

Screenshot 2:

DEVICE	TYPE	SIZE
[/dev/sda	local disk	20.000G ▶]
partition 1	new, BIOS grub spacer	1.000M ▶
partition 2	new, to be formatted as ext4, mounted at /	19.997G ▶

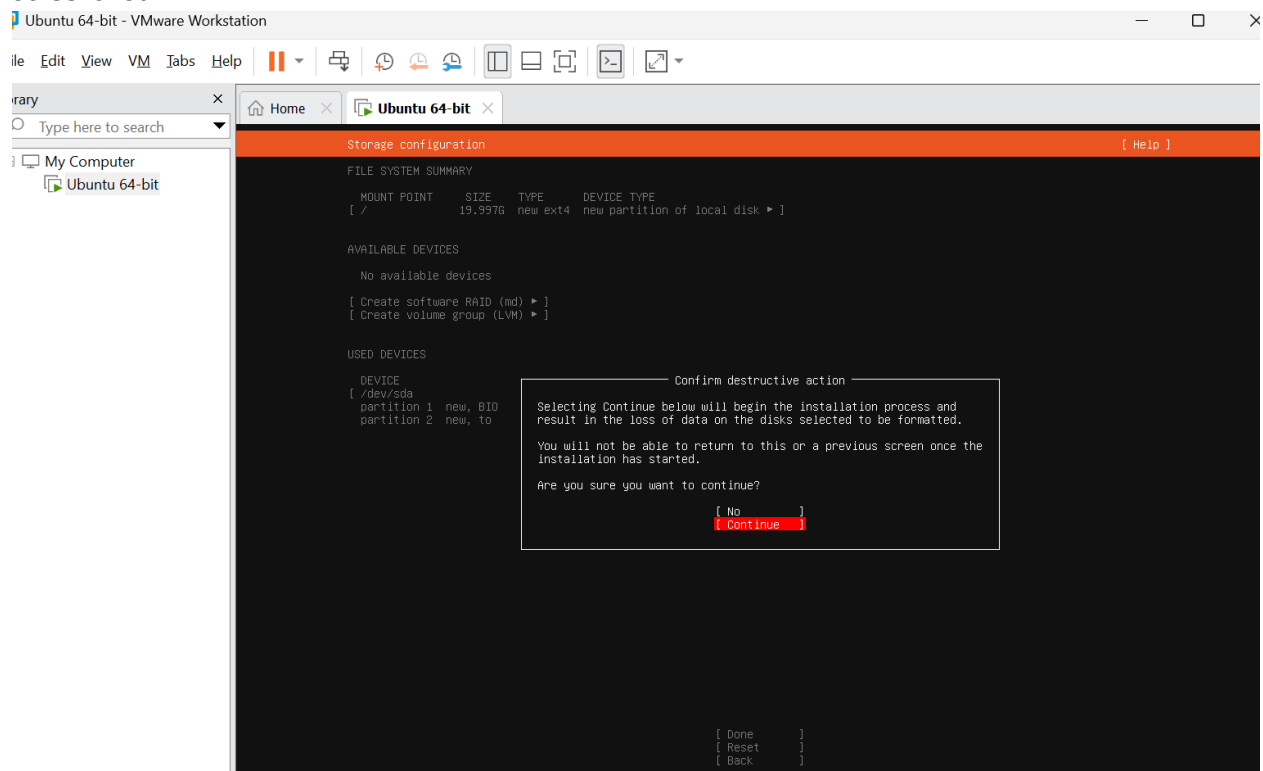
Task 11: Review the partitions

Screenshot 1:

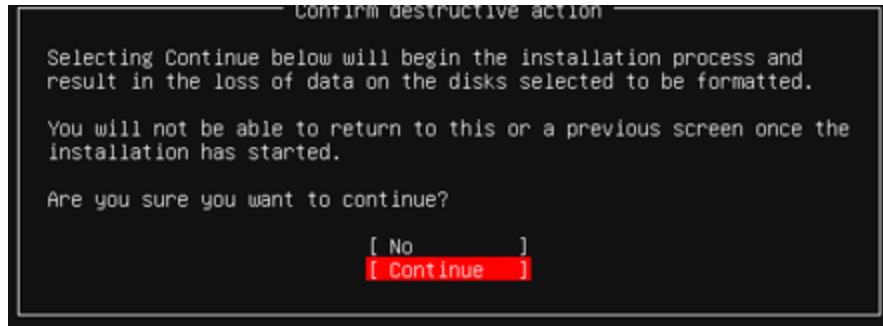


Task 12: Configure the storage changes

Screenshot1:

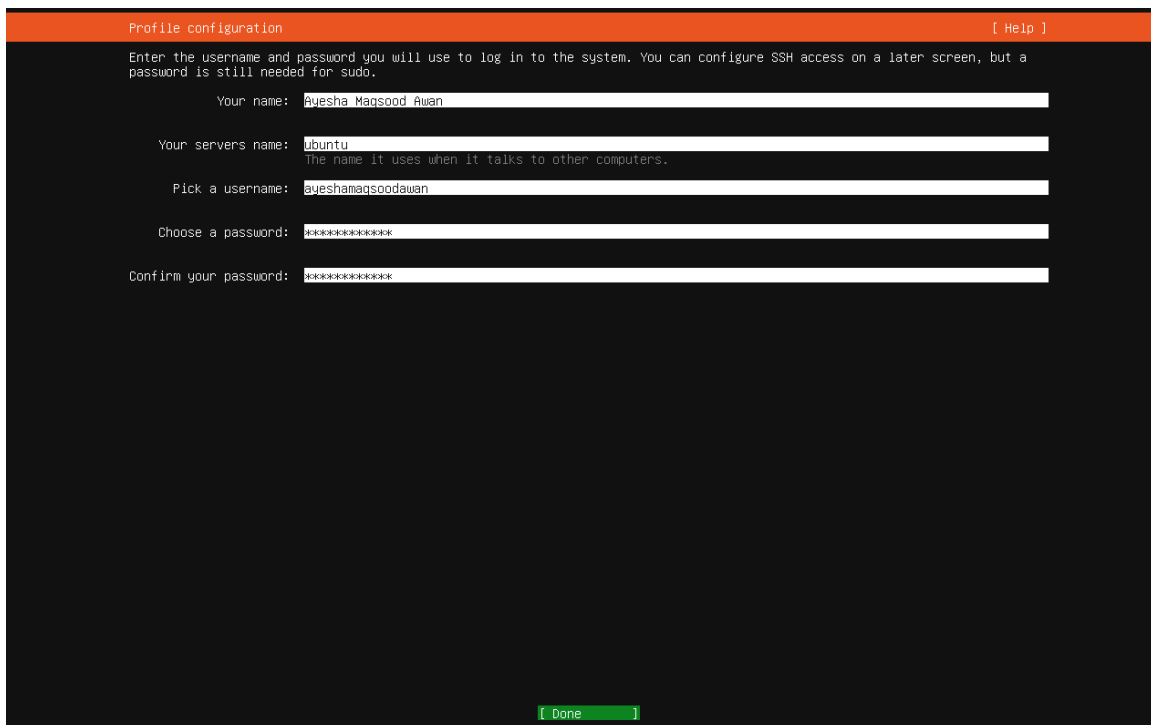


Screenshot 2: Storage changes confirmation.

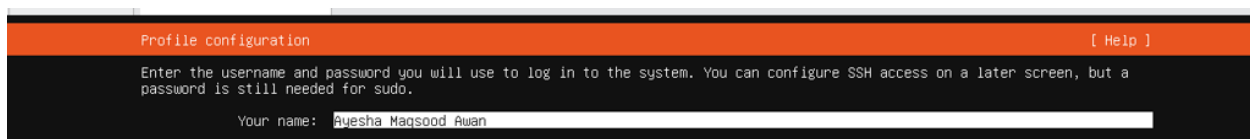


Task 13: Configure the Ubuntu profile

Screenshot 4: Complete profile setup.



Screenshot 1: Profile showing actual name



Screenshot 2: server name set to ubuntu.

Profile configuration

[Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Your servers name:

The name it uses when it talks to other computers.

Screenshot 3: username matching github username.

Pick a username:

Upgrade to Ubuntu Pro

[Help]

Upgrade this machine to Ubuntu Pro for security updates on a much wider range of packages, until 2036. Assists with FedRAMP, FIPS, STIG, HIPAA and other compliance or hardening requirements.

[About Ubuntu Pro ►]

() Enable Ubuntu Pro

(X) Skip for now

You can always enable Ubuntu Pro later using the 'pro attach' command.

[Continue]

[Back]

SSH configuration

[Help]

You can choose to install the OpenSSH server package to enable secure remote access to your server.

[X] Install OpenSSH server

[X] Allow password authentication over SSH

[Import SSH key ►]

AUTHORIZED KEYS

No authorized key

Featured server snaps

[Help]

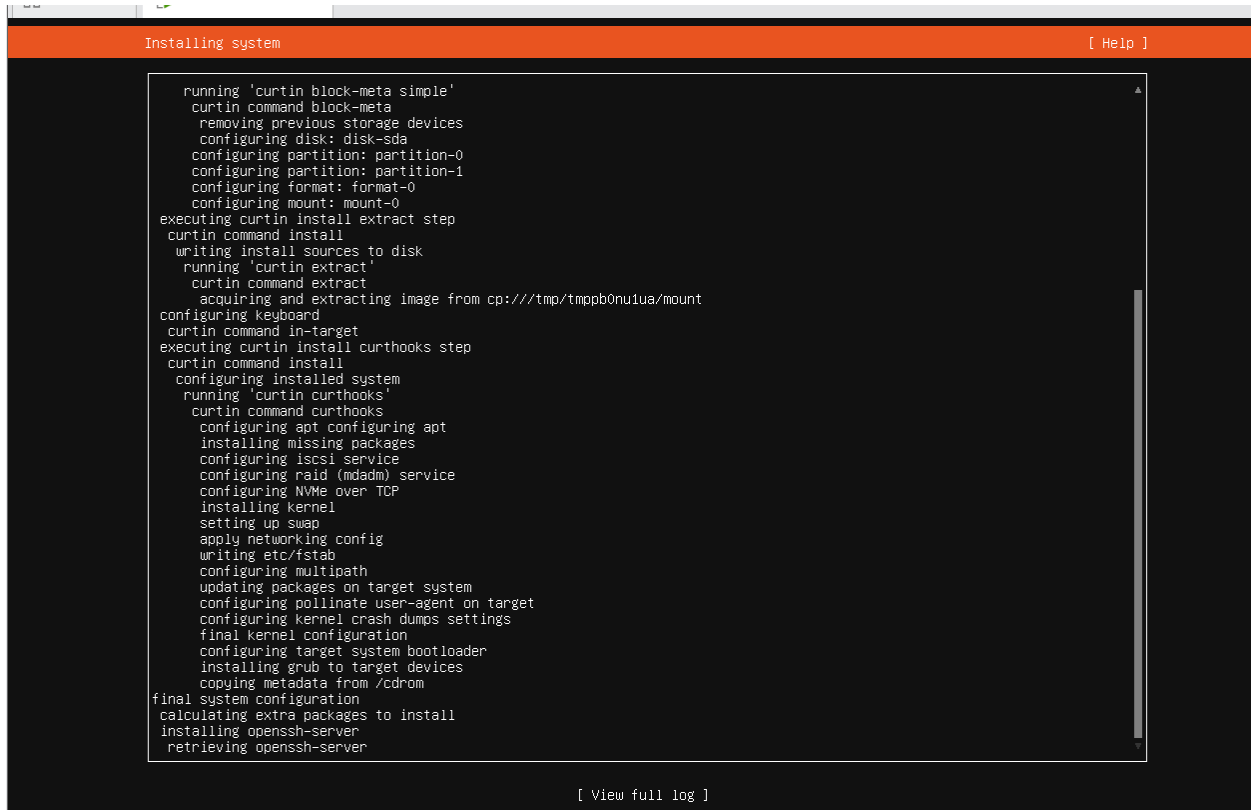
These are popular snaps in server environments. Select or deselect with SPACE, press ENTER to see more details of the package, publisher and versions available.

[] microk8s	canonical✓	Kubernetes for workstations and appliances	►
[] nextcloud	nextcloud✓	Nextcloud Server - A safe home for all your data	►
[] wekan	xet7	Open-Source kanban	►
[] canonical-livepatch	canonical✓	Canonical Livepatch Client	►
[] rocketchat-server	rocketchat✓	Rocket.Chat server	►
[] mosquitto	mosquitto✓	Eclipse Mosquitto MQTT broker	►
[] etcd	canonical✓	Resilient key-value store by CoreOS	►
[] powershell	canonical✓	PowerShell for every system!	►
[] sabnzbd	safihre	SABnzbd - The automated Usenet download tool	►
[] wormhole	snapcrafters	get things from one computer to another, safely	►
[] aws-cli	aws✓	Universal Command Line Interface for Amazon Web Services	►
[] slcli	softlayer	Python based SoftLayer API Tool.	►
[] doctl	digitalocean✓	The official DigitalOcean command line interface	►
[] keepalived	keepalived-project✓	High availability VRRP/BFD and load-balancing for Linux	►
[] prometheus	canonical✓	The Prometheus monitoring system and time series database	►
[] lxd	canonical✓	LXD - container and VM manager	►

[Done]
[Back]

Task 14: Complete the Software Installation

Screenshot 1:



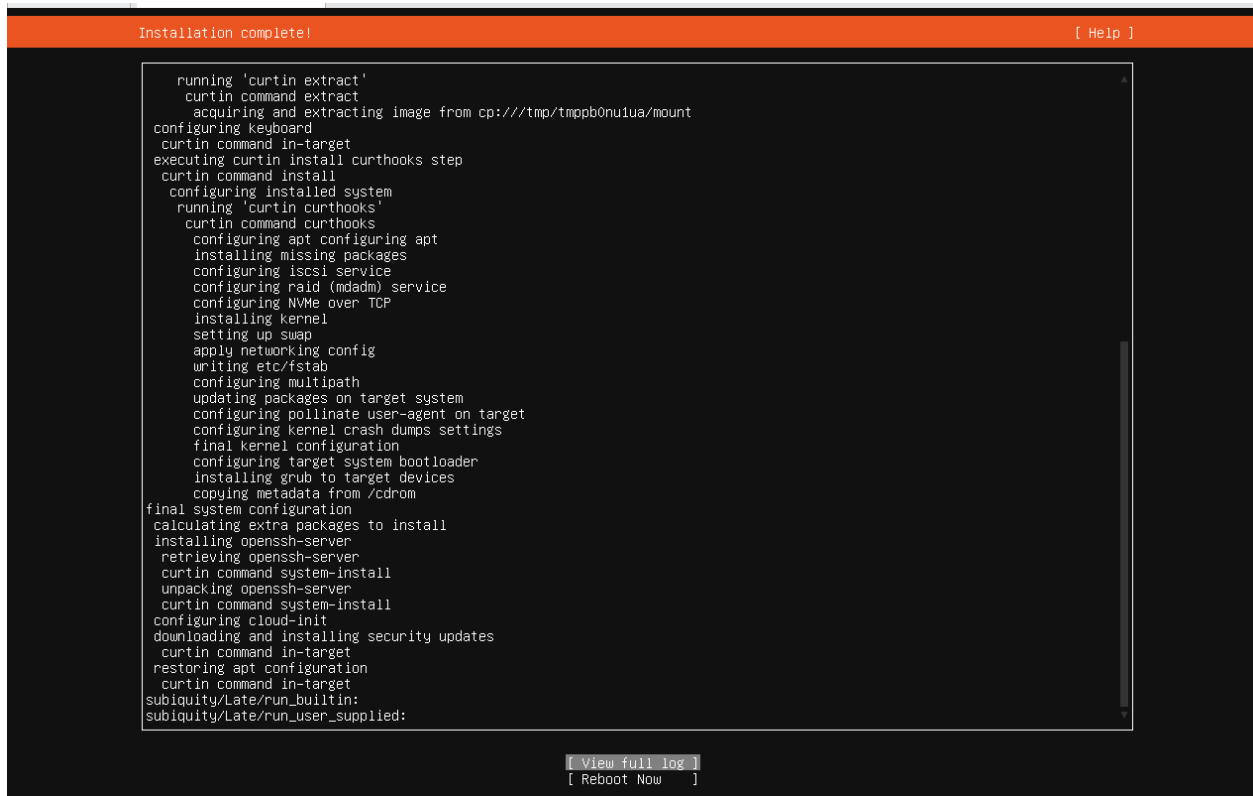
```
Installing system [ Help ]

running 'curtin block-meta simple'
curtin command block-meta
removing previous storage devices
configuring disk: disk-sda
configuring partition: partition-0
configuring partition: partition-1
configuring format: format-0
configuring mount: mount-0
executing curtin install extract step
curtin command install
writing install sources to disk
running 'curtin extract'
curtin command extract
acquiring and extracting image from cp:///tmp/tmpb0nu1ua/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
configuring installed system
running 'curtin curthooks'
curtin command curthooks
configuring apt configuring apt
installing missing packages
configuring iscsi service
configuring raid (mdadm) service
configuring NVMe over TCP
installing kernel
setting up swap
apply networking config
writing etc/fstab
configuring multipath
updating packages on target system
configuring pollinate user-agent on target
configuring kernel crash dumps settings
final kernel configuration
configuring target system bootloader
installing grub to target devices
copying metadata from /cdrom
final system configuration
calculating extra packages to install
installing openssh-server
retrieving openssh-server

[ View full log ]
```

Task 15: Reboot and verify the ubuntu identity

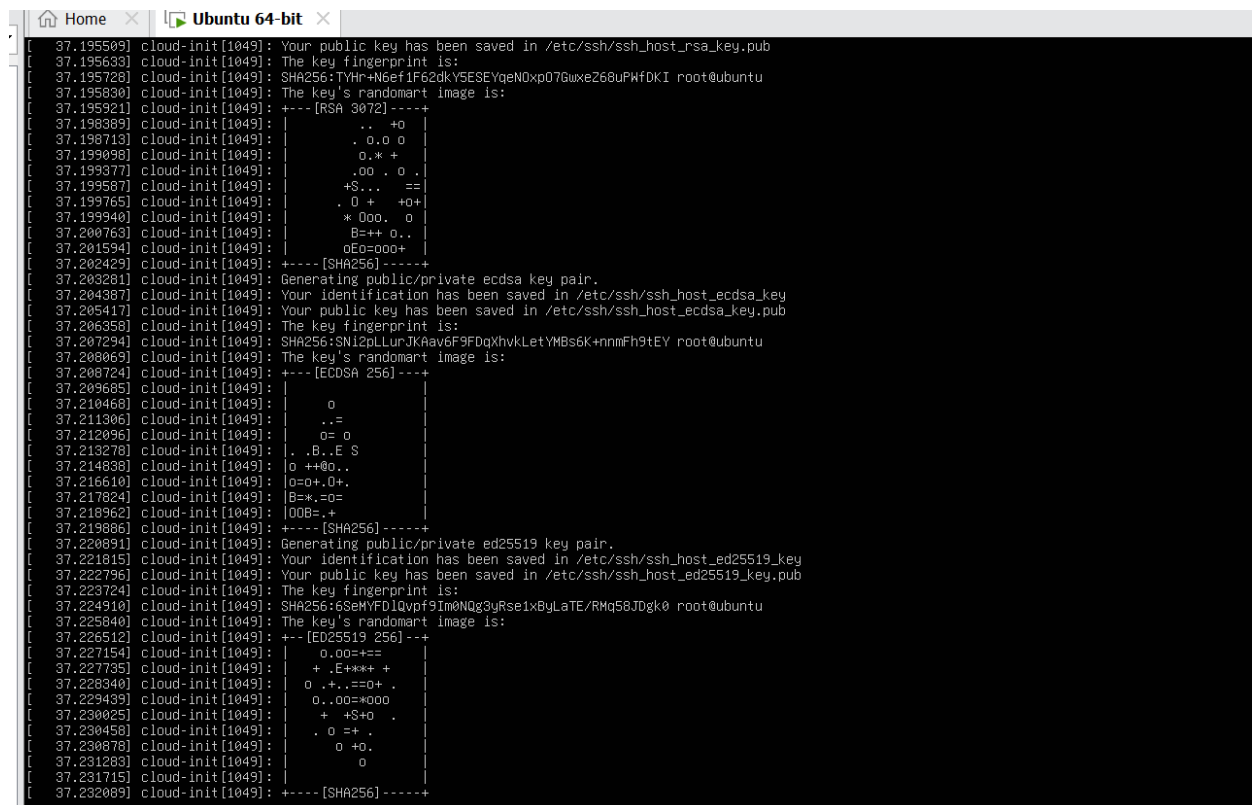
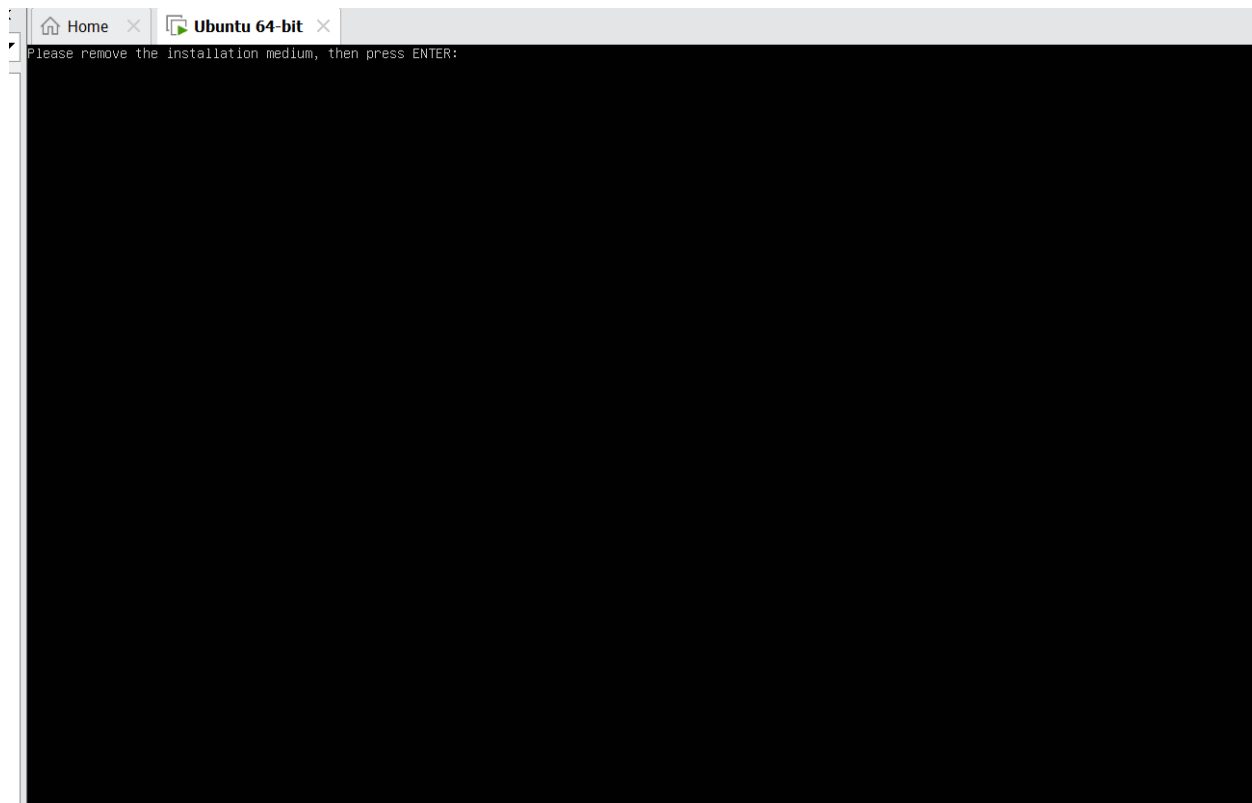
Screenshot 1:



```
Installation complete! [ Help ]

running 'curtin extract'
curtin command extract
  acquiring and extracting image from cp:///tmp/tmpbb0nu1ua/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
  configuring installed system
  running 'curtin curthooks'
  curtin command curthooks
    configuring apt configuring apt
    installing missing packages
    configuring iscsi service
    configuring raid (mdadm) service
    configuring NVMe over TCP
    installing kernel
    setting up swap
    apply networking config
    writing etc/fstab
    configuring multipath
    updating packages on target system
    configuring pollinate user-agent on target
    configuring kernel crash dumps settings
    final kernel configuration
    configuring target system bootloader
    installing grub to target devices
    copying metadata from /cdrom
final system configuration
calculating extra packages to install
installing openssh-server
  retrieving openssh-server
  curtin command system-install
  unpacking openssh-server
  curtin command system-install
configuring cloud-init
downloading and installing security updates
  curtin command in-target
restoring apt configuration
  curtin command in-target
subiquity/late/run_builtin:
subiquity/late/run_user_supplied:
```

[View full log]
[Reboot Now]



Screenshot 2: Ubuntu login screen

```
[ 37.231715] cloud-init[1049]: |
[ 37.232089] cloud-init[1049]: +----[SHA256]-----+
^^
ubuntu login: ayeshamaqsoodawan
Password:
```

Screenshot 3: Successful terminal login

```
ubuntu login: ayeshamaqsoodawan
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun Sep 13 05:18:48 PM UTC 2026

System load:  0.04          Processes:           238
Usage of /:   28.0% of 19.51GB   Users logged in:    0
Memory usage: 22%           IPv4 address for ens33: 192.168.68.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

8 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ayeshamaqsoodawan@ubuntu:~$
```

Screenshot 4: <github-username>@ubuntu prompt

```
ayeshamaqsoodawan@ubuntu:~$
```

```
ayeshamaqsoodawan@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:15:c2:1d brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx00c2915c21d
    inet 192.168.68.129/24 metric 100 brd 192.168.68.255 scope global dynamic ens33
        valid_lft 1106sec preferred_lft 1106sec
    inet6 fe80::20c:29ff:fe15:c21d/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
ayeshamaqsoodawan@ubuntu:~$
```

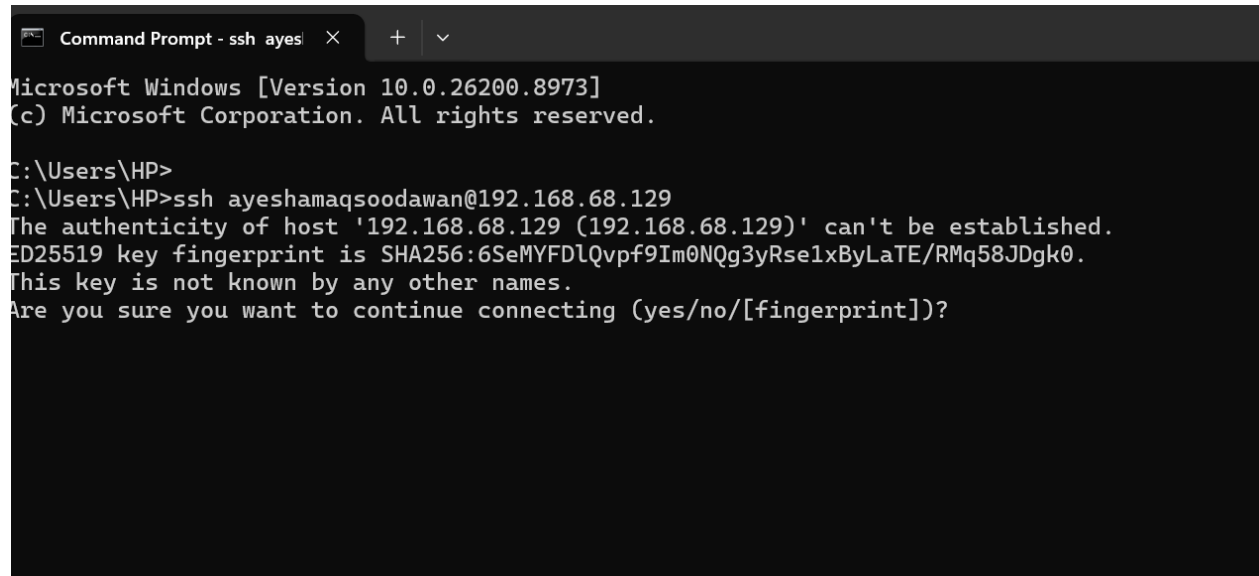
ubuntu_ip_addr_command.png

```
inet 192.168.68.129/24
```

ubuntu_ip_address.png

Task 17: Connect to ubuntu Server from windows

Screenshot 1:



```
Command Prompt - ssh ayes  X  +  v

Microsoft Windows [Version 10.0.26200.8973]
(c) Microsoft Corporation. All rights reserved.

C:\Users\HP>
C:\Users\HP>ssh ayeshamaqsoodawan@192.168.68.129
The authenticity of host '192.168.68.129 (192.168.68.129)' can't be established.
ED25519 key fingerprint is SHA256:6SeMYFDlQvpf9Im0NQg3yRse1xByLaTE/RMq58JDgk0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])?
```

Screenshot 2: Accetped SSH server fingerprint

```
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.68.129' (ED25519) to the list of known hosts.
ayeshamaqsoodawan@192.168.68.129's password:
```

Screenshot 3: Successful SSH session from windows.

```
C:\Users\HP>ssh ayeshamaqsoodawan@192.168.68.129
ayeshamaqsoodawan@192.168.68.129's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun Sep 13 05:55:18 PM UTC 2026

System load:  0.08               Processes:            209
Usage of /:   28.1% of 19.51GB   Users logged in:     0
Memory usage: 23%               IPv4 address for ens33: 192.168.68.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

8 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

Screenshot 4: Remote SSH prompt showing <github-username>@ubuntu

```
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

ayeshamaqsoodawan@ubuntu:~$
```